

# 6

## Vibrational Effects in Graphene Nanoribbons

### Contents

---

|     |                                                                                   |     |
|-----|-----------------------------------------------------------------------------------|-----|
| 6.1 | Introduction . . . . .                                                            | 120 |
| 6.2 | Coulomb-blockade Spectroscopy . . . . .                                           | 121 |
| 6.3 | Analysis of the Franck–Condon Conductance Blockade                                | 123 |
| 6.4 | Simulation of the Stability Diagram . . . . .                                     | 126 |
| 6.5 | Role of the Maleimide Groups: <i>Ab-Initio</i> and Statistical Analysis . . . . . | 128 |
| 6.6 | Concluding Remarks . . . . .                                                      | 133 |
| 6.7 | Project Overview . . . . .                                                        | 134 |

---

### 6.1 Introduction

As we saw in the last chapter, MGNRs could provide a platform to electrically manipulate electronic states at high temperatures. Since spin coherence times up to  $\mu\text{s}$  have been measured in a comparable temperature range [99, 114], circumstances appear to be ideal for using single MGNRs to exploit spin and topological phenomena for quantum applications; however, only devices with sharp features, where all the quantum states can be accessed properly, can be used for quantum experiments.

Because of the strong  $\pi-\pi$  interactions between their large conjugated  $sp^2$ -carbon

planes, MGNRs can easily aggregate in the solid state and in solution, making the integration of MGNRs into electronic nanodevices challenging. For improving liquid-phase MGNR dispersion, both physical techniques (i.e., sonication exfoliation [115]) and chemical treatment with a strong acid [224] have been used. These approaches, however, have limited dispersion effects and, often, MGNRs clusters persist.

To date, two basic techniques for producing liquid-phase-dispersible MGNRs have been proposed. The first approach involves the addition of alkyl substituents along the edges. For example, dodecyl-decorated MGNRs with lateral widths of 0.7–1.1 nm and lengths of more than 200 nm are often the go to choice [203], but strong sonications are unfortunately still necessary, which may be detrimental to the honeycomb lattice, as carbon nanotubes have indicated in the past [225, 226].

The other approach involves the addition of bulky 3D groups at the periphery of the MGNRs. The dimensions of these substituents are typically greater than the graphite interlayer distance  $\simeq 0.34$  nm, hence they should dramatically disrupt the  $\pi - \pi$  interactions, and aid the dispersion of individual ribbons in the liquid phase. A little shaking or very mild sonication in common organic solvents such as THF, dichloromethane, and toluene is reported to yield homogeneous dispersions [207].

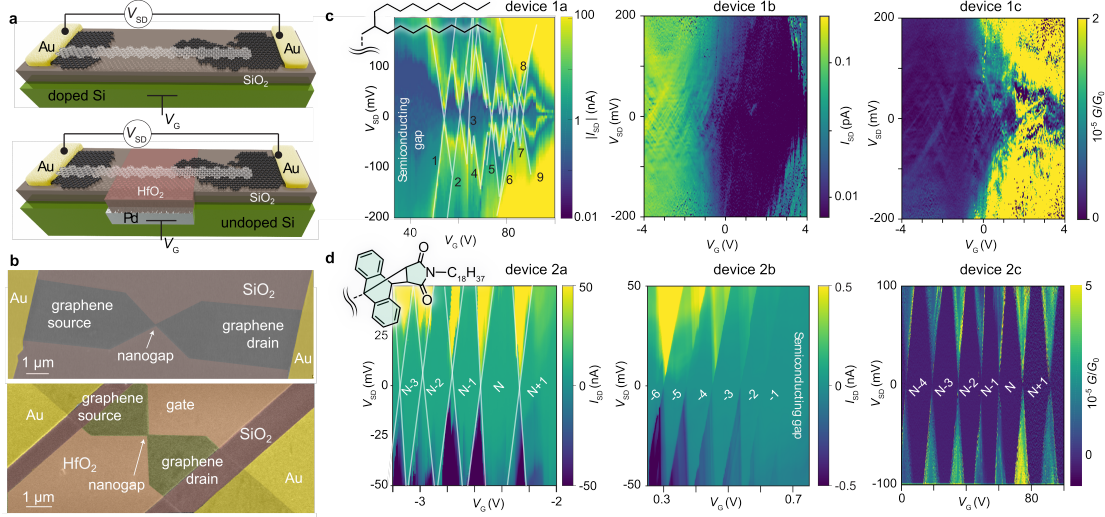
In this chapter, we will attempt to demonstrate that attaching 3D substituents with a high degree of steric hindrance along the ribbon edges can improve the solubility of graphene nanoribbons in common apolar solvents.

## 6.2 Coulomb-blockade Spectroscopy

To test the idea that bulky 3D groups increase the degree of MGNRs de-bundling in solution, we tested two nanoribbons carrying two different edge functionalisations. The first was decorated with dodecyl alkyl (MGNR1), while in the second maleimide groups were used (MGNR2) (Fig. 6.1). Furthermore, to investigate possible differences at the device performance level, we connected these ribbons to two device architectures.

The gate is provided by either an n-doped Si/SiO<sub>2</sub> backgate wafer ( $t_{\text{ox}} \simeq 300$  nm) or a local Pd electrode etched into undoped Si and covered by HfO<sub>2</sub> ( $t_{\text{ox}} \simeq 10$  nm)

(Fig. 6.1a,b). As mentioned in Ch.3, MGNRs are added by drop casting from solutions, with a yield of about 5 devices out of 100 electroburned gaps, and the existence of a nanoribbon connecting the two leads is identified as a sharp increase of the conductance  $G = \partial I_{SD}/\partial V_{SD}$  from  $G < 1 \text{ nS}$  to  $\simeq 100 \text{ nS}$ .



**Figure 6.1:** **a**, The device geometries utilised are illustrated in the diagram: an MGNR (side groups are not shown for clarity) bridges two graphene leads coupled to Au pads. The gate is either an n-doped Si wafer with a 300 nm SiO<sub>2</sub> insulating layer on top or a Pd electrode buried in undoped Si with a  $\simeq 10 \text{ nm}$ -thick HfO<sub>2</sub> layer on top (bottom). **b**, False-color scanning electron microscopy images of two typical devices for the two geometries. **c**, The source–drain current against bias and gate voltages are shown in stability diagrams for three devices derived with GNR1. **d**, Stability diagrams for three GNR2 devices, displaying the source–drain conductance in units of the conductance quantum  $G_0$ . All measurements are done below  $T = 500 \text{ mK}$ , and diamonds are labelled with regard to an arbitrary number of electrons,  $N$ , whenever possible.

According to the operating mechanism of SETs outlined in Ch. 2, electrons tunnel through a single channel, whose potential is regulated by  $V_G$  whose role is to align the MGNR levels in resonance with the leads. Because of Coulomb repulsion, an electron already present in the conduction channel would prevent the flow of additional electrons, thus suppressing the device conductance.

The high degree of bundling expected for MGNR1 made it impossible to produce SETs with only one ribbon bridging the nanogap, and the typical stability diagram shows a large region of suppressed conductance (Fig. 6.1c), corresponding to the bandgap of semiconducting MGNRs, followed by multiple overlapping Coulomb

diamonds of varying sizes. From the height of the diamonds, and knowing the ribbon width and length [216], some of the most discernible diamonds yield  $l_{\text{MGNR}} \simeq 28, 90, 40, 17, 108, 60, 50$ , and  $38 \text{ nm}$ , and no sharp features inside the diamonds could be distinguished.

In contrast, SETs produced with MGNR2 exhibit very clean conduction characteristics and a level of detail equivalent to ultra-clean CNTs (Fig. 6.1d). The diamonds are distinctly periodic, have identical slopes and diameters, and well-defined edges, indicating that each device has a single nanoribbon. Parallel to the Coulomb edges, the excited states are observable as intense, conductance lines running parallel to the diamonds. The high contact-impedances observed are likely influenced by contact lengths limited by the MGNR length [227], which is less than the transfer length<sup>†</sup>.

A zoom-in to one of the diamonds highlights the exceptional cleanliness of the features and a full spectroscopy of the conducting element: the edges of the diamonds are suppressed, and a series of multiple excited states can be clearly observed (Fig. 6.2a), with a  $\hbar\omega_\nu = (7.5 \pm 0.1) \text{ meV}$  feature common to all devices. We observed a gradual quenching of the edge-conductance suppression upon increasing  $T$  (Fig. 6.2d). Similar to the Franck–Condon principle in molecular spectroscopy, electron–vibron coupling might suppress transport for low-lying vibronic states, because electrons are preferably transmitted when the nuclei-position shifts are minimal.

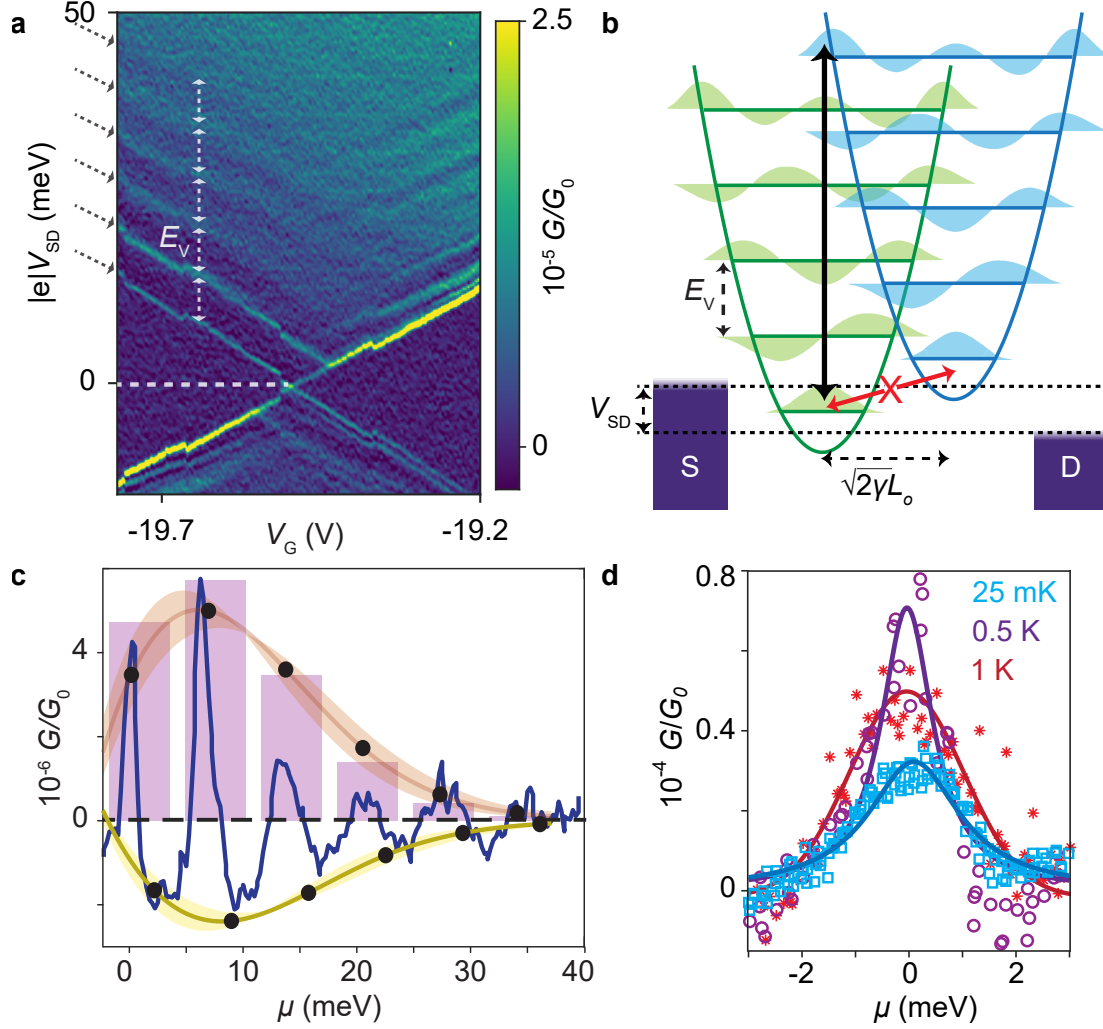
### 6.3 Analysis of the Franck–Condon Conductance Blockade

We shall now attempt to explain the suppression of conductance lines parallel to diamond edges within the Franck–Condon blockade framework.

---

<sup>†</sup>The transfer length is the distance over which a fraction equal to  $1 - e^{-1}$  of the total current is *transferred* from the contact metal to the semiconducting channel ( $e$  is the Euler’s number), and it is frequently used as a metric for quantifying the coupling between electrical contacts and conduction channel (small transfer lengths result in greater couplings, and, consequently, lower contact resistances) [228].





**Figure 6.2:** **a**, Detail of vibrational state suppression in the differential conductance  $G$ , in units of quantum conductance  $G_0$  versus  $V_{SD}$  and  $V_G$ . Arrows indicate the excited states, which we assign to vibrational given the equal spacing between them.  $T = 20$  mK. **b**, The Frank-Condon blockade in the transport properties of nanodevices is depicted schematically, with the equilibrium coordinates for two adjacent charge states (green and blue) represented horizontally. Strong vibronic coupling suppresses ground-to-ground transitions exponentially, whereas ground-to-excited-state transitions become viable at higher bias. **c**, Blockade peaks as a function of the chemical potential  $\mu$  as experimentally observed (blue line) and as expected by Frank-Condon theory for maxima (orange) and minima (yellow). Poisson histogram for the vibrational quanta (purple) is superimposed. **d**, Lifting of the Frank-Condon blockade for a transmission channel upon increasing the temperature from 25 mK (blue) to 1 K (red), together with Lorentzian fits to the data (solid lines).

According to standard electronic absorption theory, the spectrum resulting from the excitation with wavelenths within the vibrational region is a progression of absorption peaks rising separated by multiples of the vibrational quanta, and with the intensity lines of the spectrum that follows a Poisson distribution [229]. The amplitude of each peak in the vibrational progression is governed by the Franck–Condon factors, which are defined as transition probabilities, according to standard quantum theory, between the vibrational states of two different electronic states (ground and excited states) over many vibrational modes.

If we assign the excitation of vibrational states to the equispaced parallel conductance lines within a Coulomb cone, we might be able to link the Franck–Condon factor to the DC conductance maxima. Thus, by extracting and averaging within a bias window parallel lines to the Coulomb edges, we observed that the conductance peaks follows a known progression [230].

Franck–Condon theory describes the coupling electron–vibron  $\gamma$ , i.e. how an addition of an electron changes the electronic wave function and produces a shift in the equilibrium coordinate of a molecule. If  $\gamma$  is large, then charge transport may be suppressed through low-lying vibronic MGNR states (Fig. 6.2b). Furthermore, Franck–Condon theory predicts that the peak conductance of the vibronic transitions,  $G_n^{\max}$ , follows a Poisson distribution in terms of the vibron quantum number  $n$ ,

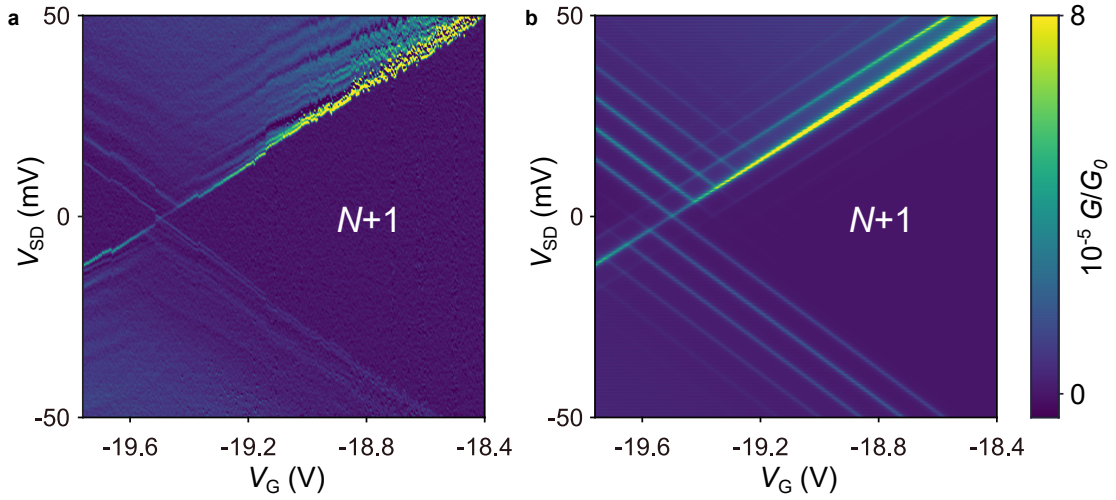
$$G_n^{\max} = \frac{\gamma^n}{n!} e^{-\gamma}, \quad (6.1)$$

By fitting the averaged conductance trace within a Coulomb cone with Eq. 6.1 we extract an electron–vibron coupling  $\gamma = 1.5 \pm 0.2$ . Fig. 6.2c shows the progression of the vibronic peaks, which arise when an electron tunnels through a molecular orbital and simultaneously excites a molecular vibrations [231].

A hint of Franck-Condon suppression apparent from Figure Fig. 6.2d where an increase in  $T$  from 25 mK to 0.5 K is followed by an increase in  $G$ . A further increase to 1 K decreases  $G$  due to significant temperature broadening. The same figure was used to calculate the electrode–MGNR coupling energy  $\Gamma \simeq (185 \pm 30) \mu\text{eV}$ , in line with the estimate detailed in the previous chapter.

## 6.4 Simulation of the Stability Diagram

We shall now try to validate our conclusions exposed in the previous section about the possibility of having a Franck–Condon suppression of the conductance by reproducing the experimental observation with a simulated model. We shall use, to this end, the model exposed in Sec. 2.4, and assume that the molecular energy level,  $\mu = C_S V_{SD} - C_G V_G$ , shifts with both bias and gate voltages.



**Figure 6.3:** Maps of the conductance for the same transition as in Fig. 6.2a, plotted on identical colour scale. **a**, Experimental conductance map for the same charge transition of Fig. 6.2a. **b**, Simulated conductance using Eq. 2.9–2.16 and the parameters from Tab. 6.1. Integrals were evaluated using the Simpson’s rule as implemented in Numpy.

The calculated conductance  $G$  (Tab. 6.1 reports the simulation parameters) is plotted as a function of  $V_{SD}$  and  $V_G$ , and is displayed next to an experimental stability diagram in Fig. 6.3. The calculated and experimental stability diagrams match each other well. Importantly, the calculated diagram shows that the single-mode–environment model reproduces the main features of our experimental data, i.e. conductance lines equally spaced by  $E_v \simeq 7$  meV, a significant contribution from the phonon bath and conductance asymmetry with bias. The calculated stability diagram contains additional conductance sidebands in the blocked regions that are not present in the experimental diagram.

**Table 6.1:** Simulation parameter for the vibrational electron transport model

| $T$<br>(mK) | $\hbar\omega_q$<br>(meV) | $\hbar\omega_c$<br>(meV) | $\hbar g_q$<br>(meV) | $\lambda$<br>(meV) | $\Gamma_S$<br>(meV)  | $\Gamma_D$<br>(meV) | $C_S$<br>(fF) | $C_G$<br>(fF)         |
|-------------|--------------------------|--------------------------|----------------------|--------------------|----------------------|---------------------|---------------|-----------------------|
| 25          | 7                        | 8.5                      | 20                   | 150                | $1.2 \times 10^{-3}$ | 0.5                 | 0.54          | $2.25 \times 10^{-2}$ |

Using the rate-equation model discussed in Section 2.4.1 for a single-phonon mode  $\hbar\omega_q = 7$  meV and a superohmic phonon bath, we calculated a SD (Fig. 6.4). Using Eq. 2.9–2.16 we reproduced our experimental data by using the set of parameters reported in Table 6.1. The calculated SD exhibit the same regularly spaced conductance peaks as observed experimentally. These result from the strongly coupled single-mode that generates multiple peaks in the DOS at low energies (Fig. 6.4c, d). At higher bias the background conductance increases due to the coupling with an increasing number of phonon modes in the bath. Moreover, the strong  $V_{SD}$  asymmetry is caused by  $\Gamma_D \gg \Gamma_S$  and may be caused by larger overlap between the MGNR and the drain than the source lead. The model does not take into account the slanted conductance lines originating from the leads and does therefore not reproduce these.

All vibrons at frequencies  $\omega_\nu$  and the associated electron–vibration coupling strengths  $g_\nu$  might be included into a quantum-mechanical description of the electron-transfer rates through the MGNR DOS [149]. The outer-sphere background, produced by the weak coupling of the MGNR with the substrate, can also be included by adding a super-Ohmic spectral density with an exponential cut-off [232]. The data agreed extremely well with the simulation, and the entire current map was faithfully reproduced (Fig. 6.3). Since the observed vibrational frequency  $\omega_\nu$  and the associated quantization energy  $E_\nu = \hbar\omega_\nu$  are much higher than the modes of a 300 nm-long vibrating CNT, indicating coupling to different vibrations. The electronic states of the MGNR are largely edge-localised [233], and can overlap and couple with the edge vibrational modes.

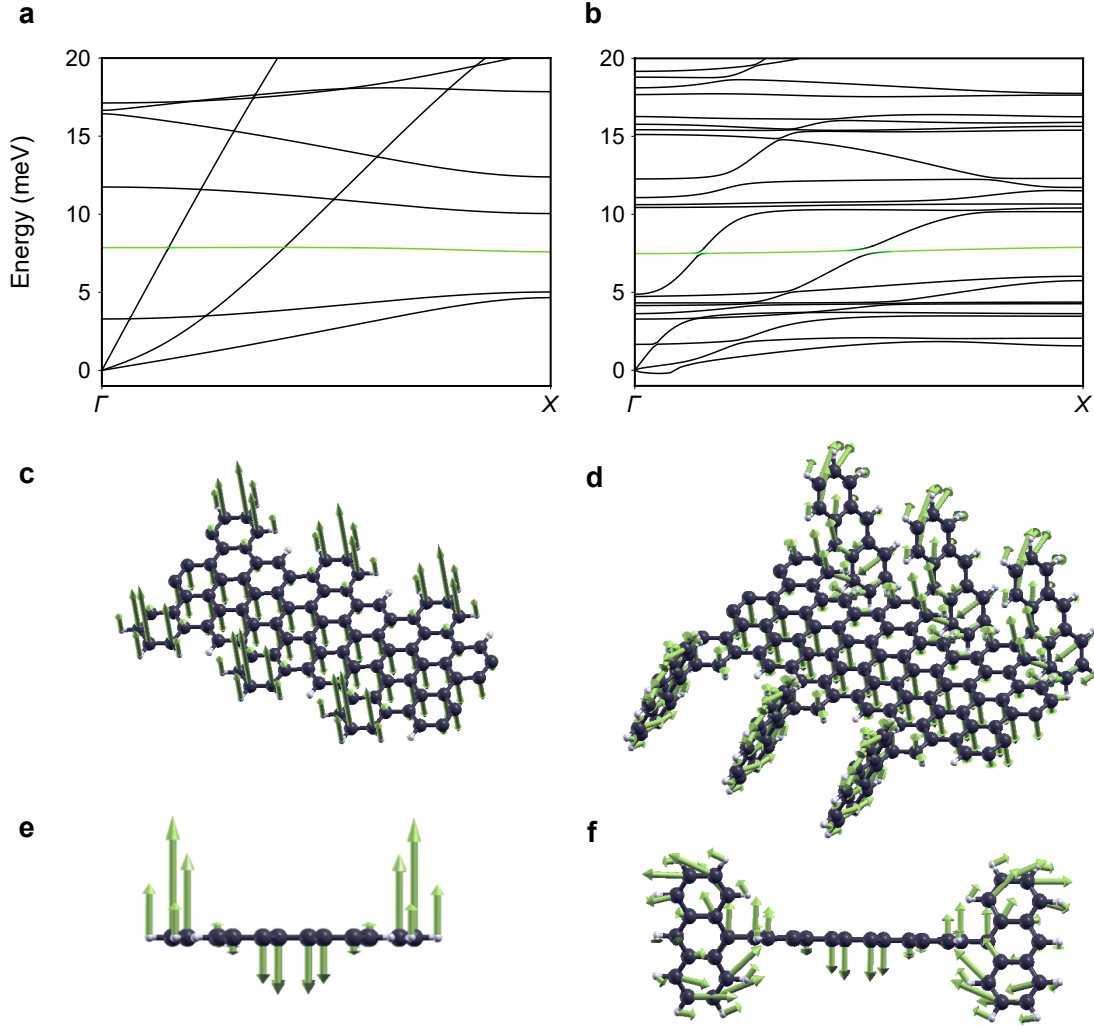
## 6.5 Role of the Maleimide Groups: *Ab-Initio* and Statistical Analysis

Since our MGNRs are synthesised with molecular precision through bottom-up synthesis, we deem appropriate to calculate from *ab-initio* the vibrational properties of a simplified MGNR structure and to attempt elucidating the origin of the Franck–Condon blockade and the nature of the vibrational mode we identified through the conductance measurements.

DFT calculations were performed using the GGA–PBE functional as implemented in the Siesta package [234, 235]. One structure was calculated with edges terminated by hydrogen atoms, another with anthracene groups, and the third with maleimide groups (Fig. 6.4 and Fig. 6.5). Along the non-periodic directions an adequate number of vacuum states were added in order to avoid unwanted interactions. An energy cut-off of 400 Ry was used together with a  $20 \times 1 \times 1$  Monkhorst–Pack grid constructed along the three Cartesian directions. The structure was optimized until the maximum force on the atoms was less than  $0.01 \text{ eV } \text{\AA}^{-1}$ . The dispersion relation is calculated with a supercell size of  $3 \times 1 \times 1$ , where 3 represents 3 unit cells along the periodic direction, and atomic displacements of the  $\Gamma$  phonon mode are visualised with the software XCrySDen. The phonon disperions were calculated using the same numerical constraints.

Firstly, we calculated the effect the anthracene groups have on the atom displacements compared to the base case (Fig. 6.4). We can qualitatively assess the emergence of new phonon modes, the majority of which are exclusive to the anthracene groups, and a few of which correlate to the MGNR backbone. In addition, the three acoustic phonons combine with optical phonons, resulting in the constant dispersion of new hybrid phonons throughout the Brillouin zone. Particularly, the dispersion relation reveals that the phonon mode at  $\hbar\omega_\nu = 7.50 \text{ meV}$  is robust at the  $\Gamma$  point and experiences mode mixing only near the zone boundary, which may be caused by strong electron–phonon interaction [236].

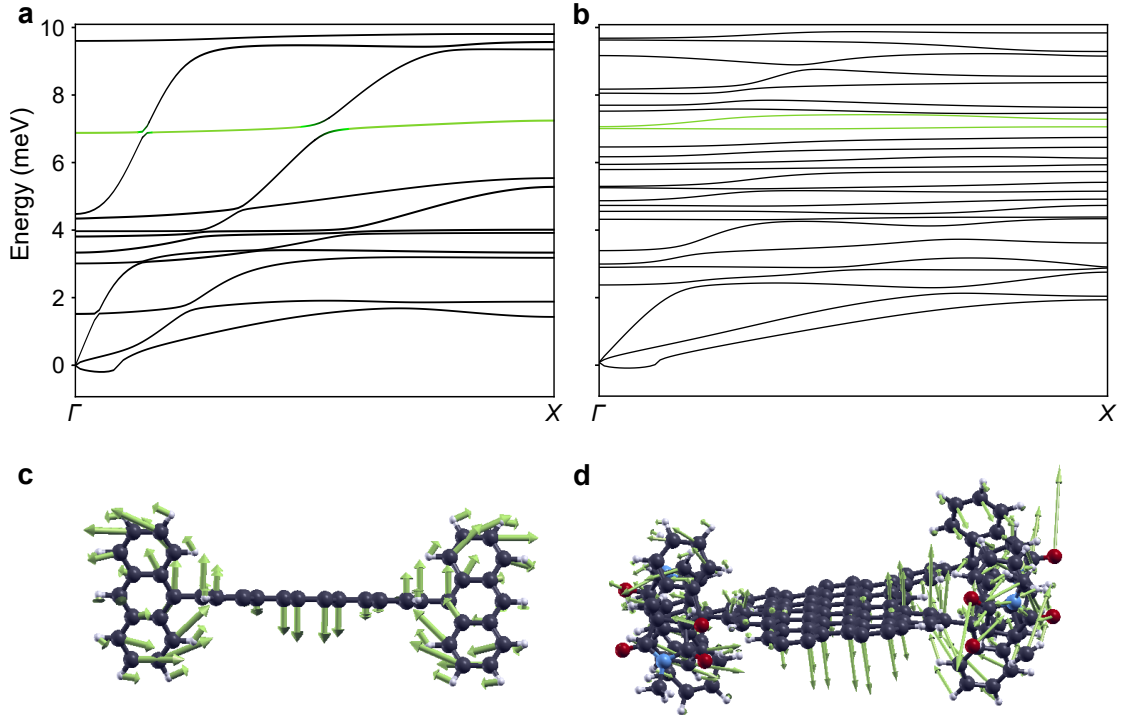
In addition, at the  $\Gamma$  point one mode is reduced from 7.86 meV to 7.48 meV, matching the experimental value (Sec. 6.3) much more accurately. We conclude,



**Figure 6.4:** **a, b**, Energy dispersion of the lowest vibrational modes when anthracene groups are attached to the edges, with the one at 7.5 meV highlighted in green. **c, d**, Relative displacement of the atoms for the 7.5 meV vibrational mode at  $\Gamma$  point. **e, f**, Side view of the relative displacement for the 7.5 meV vibrational mode at  $\Gamma$  point.

using the visualisation tool, that this phonon mode may be responsible for the MGNR backbone bending as well as the quenching of the amplitude oscillation of the atoms along the edges.

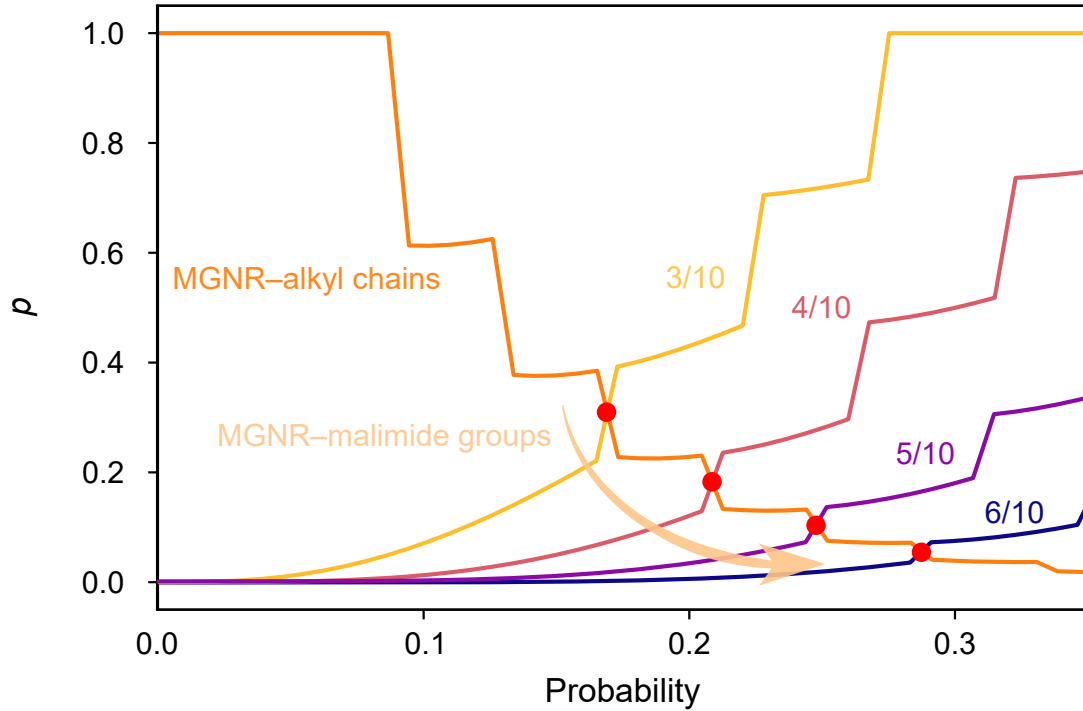
Secondly, we looked at how the inclusion of maleimide groups on the phonon dispersion following the introduction of the maleimide groups (Fig. 6.5). Compared to the anthracene, the acoustic mode at  $\hbar\omega = 7.48$  meV induce a stronger ribbon's backbone bending (Fig. 6.5d), as also put in evidence by the greater mode



**Figure 6.5:** **a, b,** Energy dispersion of the lowest vibrational modes when maleimide groups are attached to the edges, with the one at 7.5 meV highlighted in green. **c, d,** Side view of the relative displacement of the atoms for the 7.5 meV vibrational mode at  $\Gamma$  point.

amplitude. Furthermore, we note that the displacement of core-backbone atoms remains approximately constant, whereas cove-edge atoms become more rigid. This qualitative observation seems to suggest that bending will be primarily confined to the backbone centre when 3D side groups are introduced.

Despite being rigorous, our analysis may be incomplete. In fact, while Density Functional Theory (DFT) is commonly regarded as the *de facto* standard for calculating molecular and physical properties in a reasonable amount of time, issues sometime arise when accurate estimates are required. Because our system includes not only the ribbon but also the graphene leads, the question of whether maleimide groups promotes or not the dispersability of MGNRs in a solvent, thus boosting the chances to have ultra-clean SETs may still arise. In order to answer this question, we will use a simple yet powerful idea from standard statistics.



**Figure 6.6:**  $p$ -value as a function of the likelihood of obtaining a SET (probability, for short) when using MGNR2. The numerical figures represent the number of observed SETs relative to the overall number of devices cooled down in the fridge. The  $p$ -value decreases by more than an order of magnitude as the number of observed SETs increases by 2 units.

In order to exclude the possibility that the observed number of SETs (Fig. 6.1) are produced merely by chance, we shall use statistics to test if out of the total number of devices the SET formation probabilities when either MGNR1 or MGNR2 are employed are generated from the same statistical process (null hypothesis,  $H_0$ ). By setting the  $p$ -value threshold at the common value  $p = 0.05$  means that we have a 5% chance of falsely concluding that 3D bulky groups do not promote the formation of ultra-clean SETs, meaning that the SET formation probability are generated by the same statistical process.

We made one chip by depositing MGNR1 on top of GTJs and three by dropcasting MGNR2. We then wirebonded 10 devices on each chip and cooled them down at 25 mK. We counted the number of SETs for each chip by measuring high-resolution full-range stability maps and selecting whether or not to include a device based on visual inspection.



**Table 6.2:** Statistics of the SET formation probability of all the measured devices.

| MGNR Type                  | 1    | 2     | 2     | 2     |
|----------------------------|------|-------|-------|-------|
| Number of Devices Cooldown | 10   | 10    | 10    | 10    |
| Number of SETs per chip    | 0/10 | 3/10  | 4/10  | 6/10  |
| $p$ value                  | —    | 0.307 | 0.182 | 0.056 |

Assuming we can pick up a SET with equal chance by dropcasting either MGNR1 or MGNR2, we can run a binomial test to get the  $p$ -value of obtaining 0/10 SETs when MGNR1 is dropcasted. The results are stored in Tab. 6.2 and shown in Fig. 6.6.

These results seems to suggest that we cannot exclude at the 95% confidence level that the SET produced when using the 3D bulky groups are produce by chance. A couple of precisations are, however, necessary. Firstly, statistical validity of are such only in the case the experiment is repeated in the exact same conditions. Here, however, the number of observed SET across several chips cooled down is given by the total number of devices surviving the thermal shock from going to 50 K down to cryogenic temperature. This in fact does not exclude that more SET could have been actually formed but could not been measured simply due to this secondary process interfering with our statsitics.

These findings appear to imply that, at the 95% confidence level, we cannot rule out the possibility that the SET obtained when using the 3D bulky groups was produced by chance. However, a number of specifics are required. To begin, statistical validity exists only if the experiment is repeated under same conditions. However, in this case, the number of detected SETs across numerous cooled chips is determined by the total number of devices surviving the thermal shock from 50 K to cryogenic temperature. This does not rule out the possibility that more SETs were generated but could not be measured due to this secondary process interfering with our statistics.

Secondly, the number of data collected may be not sufficient for an accurate and precise statistical analysis. In particular, small samples (number of data is less than 10) tend to underestimate the standard deviation of the population, as well as having means frequently far from the true mean [237, 238]. Therefore,

at this stage we cannot exclude the role the bulky groups have on obtaining ultra-clean single-electron transistors.

Second, the amount of data collected may be insufficient for a thorough statistical analysis. Small samples (fewer than 10) in particular tend to underestimate the population standard deviation, as well as having means that are frequently distant from the true (population) mean [237, 238]. As a result, we cannot exclude the hypothesis that maleimide groups are essential to fabricate ultra-clean SETs, but also we cannot confirm it from a statistical standpoint at this time. Future study may employ a cryogenic multiplexer to increase the number of devices measurable with a single cooldown by an order of magnitude, expanding the sample size and providing more robust and accurate statistics.

## 6.6 Concluding Remarks

When compared to  $\gamma$  values reported for in-situ generated suspended CNTs, our findings are lower for some but higher for others (Tab. 6.3), and suggest enhanced electron–vibron coupling in graphene nanoribbons [239].

Remarkably, our values are equivalent to those obtained in molecular junctions comprising  $C_{140}$ , but far greater than those observed in base  $C_{60}$ . We can hypothesise that low-energy modes in the substrate’s phonon bath may facilitate transport near the diamond’s edges, so suppressing the true magnitude of the Franck–Condon effect. Decoupling the nanoribbon from the substrate environment via suspension should therefore result in higher  $\gamma$  values than for nanotubes [240].

Furthermore, first-principles studies have suggested the existence of edge-localized nonlinear modes in armchair graphene nanoribbons (AGNRs) with energies 6.8–8.7 meV [233], which would encompass our experimental and DFT findings. Based on these, the electronic wave function of our GNRs should be edge-localised and may have a non-zero overlap with edge-localised vibrational states. The corresponding vibrons are predicted to be coupled more strongly [245], which might explain why  $G \sim \hbar\omega_\nu \simeq 7$  meV is higher than lower-lying vibrons.

**Table 6.3:** Electron–vibron coupling has been described in the literature for a variety of devices.

| Material type                        | $\gamma$      | Reference |
|--------------------------------------|---------------|-----------|
| C <sub>60</sub>                      | $\ll 1$       | [241]     |
| CNT                                  | 0.5–1.1       | [240]     |
| C <sub>140</sub>                     | $\simeq 1.4$  | [242]     |
| MGNR                                 | $1.5 \pm 0.2$ | This work |
| C <sub>60</sub> (graphene junctions) | 3             | [243]     |
| CNT                                  | 3.3           | [230]     |
| Fe <sub>4</sub> SMM                  | 4.2           | [244]     |

As we saw in Ch. 1, the nanotube electronic structure is heavily dependent on the tube chirality, length, and radius, which are difficult to control and may greatly vary across devices. On the other hand, MGNRs can now be engineered with molecular accuracy, thus allowing the production of devices with more uniform performance. Currently, one of the biggest challenge is the synthesis of nanoribbons of with length spanning several hundreds of nanometer, which should lead to a higher control when positioned on top of devices.

## 6.7 Project Overview

MGNRs<sup>1</sup> were supplied by the group of Akimitsu Narita, whereas MGNRs<sup>2</sup> were synthesised in Dresden by the group of Xinliang Feng. The Franck–Condon analysis was undertaken by Dr Simen Sopp (see also [223]), whereas I acquired temperature data and conducted the corresponding analysis. Fanmiao Kong was responsible for the phonon calculations. Dr Alex Gee supported me in fabricating the devices when I was compelled to stay out of the lab for more than a month owing to covid-19 infection and subsequent health issues that necessitated medical care.

# 7

## Chemical Doping of Graphene Nanostructures

### Contents

---

|     |                                                                                                    |     |
|-----|----------------------------------------------------------------------------------------------------|-----|
| 7.1 | Introduction . . . . .                                                                             | 135 |
| 7.2 | Computational Investigation of Substitutional Doping<br>Along the Edges . . . . .                  | 136 |
| 7.3 | Electrical Transport on Methox-Edge Decorated MGNRs                                                | 141 |
| 7.4 | Substitutional Doping Through Heteroatoms Embed-<br>ding in Graphene Nanoribbons Lattice . . . . . | 142 |
| 7.5 | Concluding Remarks . . . . .                                                                       | 152 |
| 7.6 | Project Overview . . . . .                                                                         | 152 |

---

### 7.1 Introduction

When atoms with different number of valence electrons are placed into the honeycomb lattice of graphene nanoribbons, a charge transfer occurs based on the relative position of the graphene nanoribbons' Fermi level with the HOMO and LUMO of the dopant atom. For example, if a dopant's HOMO is greater than the Fermi level of the ribbon, a charge transfer from the dopant to the ribbon occurs, and the dopant acts as a donor. On the other hand, when the LUMO is less than the Fermi level of GNR, a charge transfer in the opposite direction

occurs, and the dopant acts as an acceptor.

Graphene can be doped to be p-type or n-type through chemical strategies by adsorbing dopant atoms onto the honeycomb lattice. When graphene is p-doped, the Dirac points are elevated above the Fermi level, whereas n-type doping lowers the Dirac points. In general, molecules having electron-withdrawing groups result in p-type doping, while molecules with donating groups result in n-type doping.

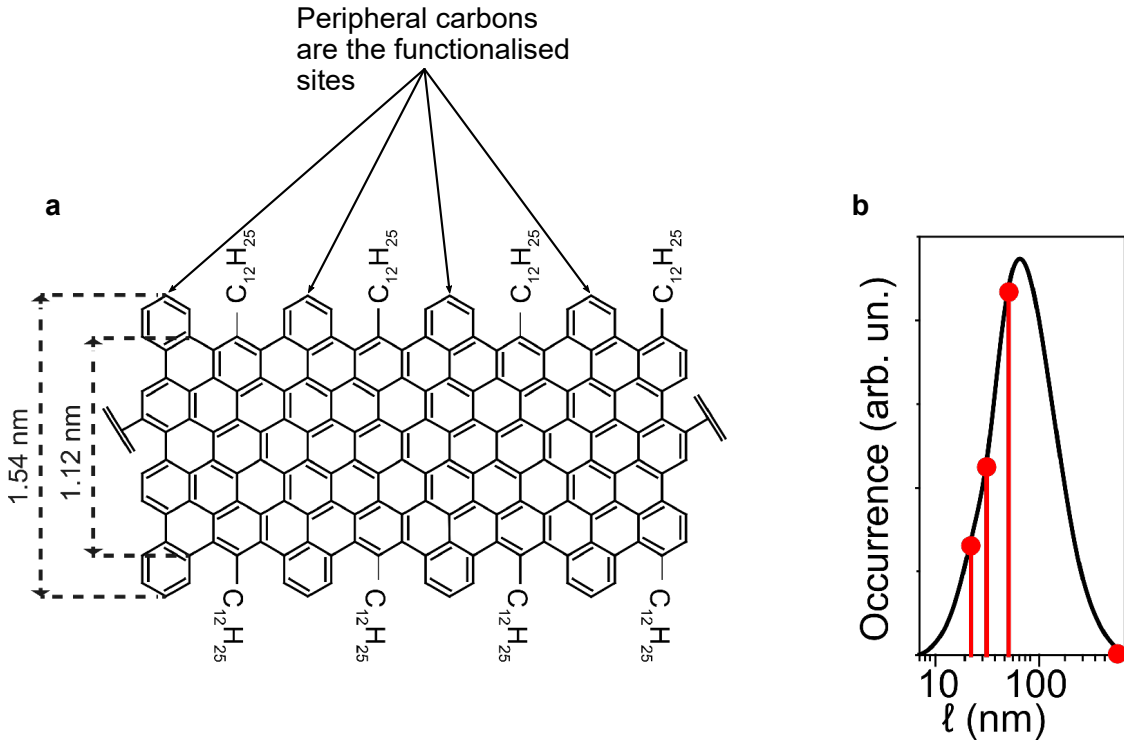
One of the most appealing aspects of graphene nanoribbons is that, thanks to bottom-up synthesis, substitutional doping, i.e. the substitution of carbon atoms with atoms carrying a different number of valence electrons, would enable chemically controlling electronic properties at a very fundamental level, overcoming a major limitation of solid-state semiconductors.

This chapter examines the impact of introducing several chemical functionalisations from both a theoretical and experimental standpoint. Specifically, DFT simulations will be used to analyse different functional groups and make predictions regarding the ensuing electrical characteristics. Then, we will examine a completely innovative method for introducing chemical functionalisations within MGNRs, namely the insertion of porphyrin units bearing transition-metal ions via the formation of heterojunctions.

## 7.2 Computational Investigation of Substitutional Doping Along the Edges

Collaborators from the Feng group in Dresden and the Müllen group in Mainz provided graphene nanoribbons with different functional groups along the edges. These ribbons share the same aromatic core and have widths ranging from  $w_{\text{GNR}} \simeq 1.1\text{--}1.2\text{ nm}$  with lengths which follow a gaussian distribution (Fig. 7.1). This last point comes from that the separation between ribbons having different length is carried out by size-exclusion chromatography means, where the average residence time in the pores created by the column packing depends on the effective size of the analyte molecules. GNRs larger than the average pore size of the packing are excluded and thus suffer essentially no retention; such species are the first to be eluted and

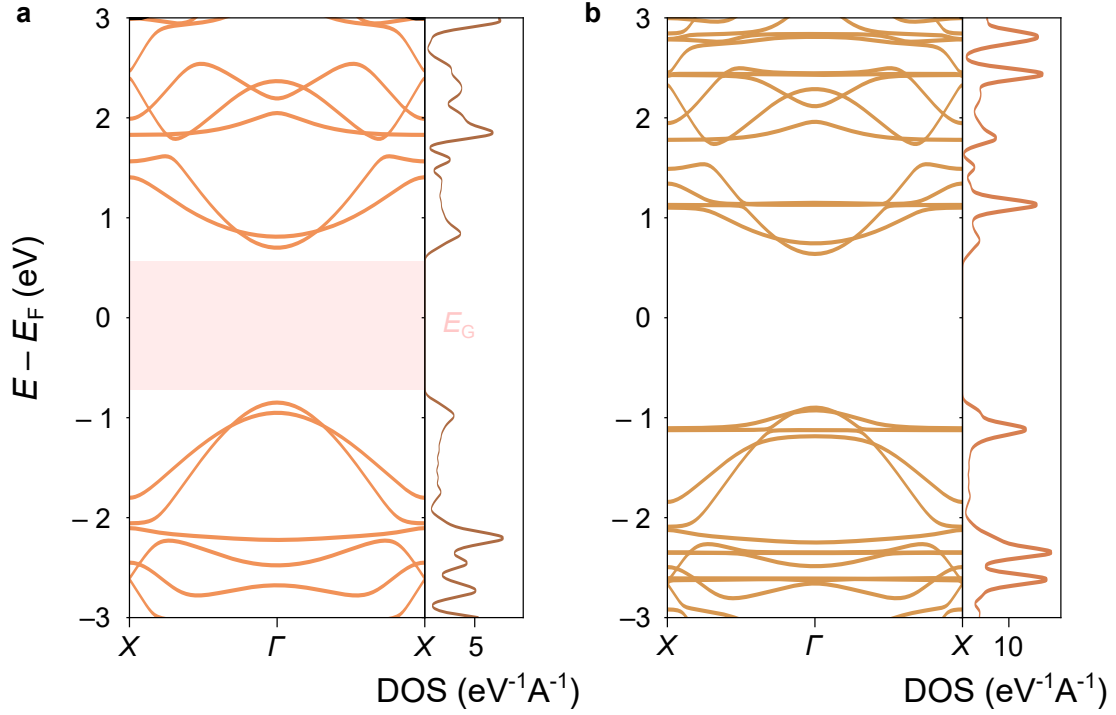
are ribbons with longer lengths. On the other hand, ribbons having diameters significantly smaller than the pores may penetrate or permeate throughout the pore network and are thus retained inside the column for the greatest time.



**Figure 7.1:** **a**, Structure of a 4-Cove graphene nanoribbon. Highlighted are the carbon sites where dopant atoms are introduced through chemical synthesis. **b**, Distribution of MGNR lengths  $l$  as obtained from size-exclusion chromatography on the polymerized molecular precursor (black). Red dots and vertical lines identify various length obtainable with different elution curves. Adapted from [246].

Unfortunately these ribbon featured very poor solubility, even after several sonication and dilution cycles in most of the common organic solvents. Since it was not possible to carry out a thorough experimental study at the device level, we have investigated the structural and electronic properties of GNR–H and compared them with the ones of GNR–OMe, GNR–Br, GNR–Cl, and GNR–Anthracene through DFT simulations.

Ab-initio calculations were performed by using the local density approximation (LDA)-CA and the generalised-gradient approximation (GGA)-PBE exchange–correlation functionals as implemented in Siesta [234, 235]. To simplify the geometry,



**Figure 7.2:** **a**, Band structure and density of states for a 4-Cove GNR-H and **b**, a 4-Cove GNR-Anthracene calculated with the exchange–correlation functional GGA-PBE.

alkyl chains were removed and replaced with hydrogen atoms. Sufficient vacuum distance is added along the non-periodic directions to avoid unwanted interactions. Numerical atomic pseudopotential was used together with an energy cut-off of 400 Ry. Monkhorst Pack k-point-grid was chosen to be  $(20 \times 1 \times 1)$  along the three Cartesian directions. The structure is optimized until the maximum force on the atoms is less than  $0.01 \text{ eV \AA}^{-1}$ .

Fig. 7.2 shows the comparison between the base case and the same ribbon functionalised with anthracene along the edges. Compared to the base case, the introduction of anthracene groups introduces sharp features in the DOS in correspondence of flat lines in the band structure, which may be attributable to a more kind of discrete spectrum. This observation is in agreement with predictions at the NEGF level and known as even–odd conductance effect [247]. In particular, for functional groups with an odd number of rings, the local density of states resides on the site of the guest molecule which is bound covalently to the nanoribbon,